



FINAL MARK

GIRRAWEE HIGH SCHOOL
MATHEMATICS
HALF YEARLY EXAMINATION
2014
ANSWERS COVER SHEET

Name: _____

QUESTION	MARK	H2	H3	H4	H5	H6	H7	H8	H9
1-3	/3	√			√				√
4	/1	√							√
5,6	/2	√	√		√				√
7	/1	√			√				√
8,9	/2	√	√		√				√
10	/1	√	√						√
Total M.C.	/10								
11	/15	√	√						√
	/15								
12	/15	√			√				√
	/15								
13a	/3	√							√
b	/5	√						√	√
c	/3	√	√		√				√
d	/7	√		√	√				√
	/18								
14ab	/7	√						√	√
c	/4	√	√					√	√
d	/4	√	√		√				
	/15								
15a	/4	√	√		√				√
b	/11	√	√		√	√			√
	/15								
16a	/8	√		√	√				√
b	/4				√				
	/12								
TOTALS	/100	/100	/46	/15	/64	/11		/16	/100

- H1 seeks to apply mathematical techniques to problems in a wide range of practical contexts.
- H2 constructs arguments to prove and justify results.
- H3 manipulates algebraic expressions involving logarithmic and exponential functions.
- H4 expresses practical problems in mathematical terms based on simple given models.
- H5 applies appropriate techniques from the study of calculus, geometry, probability, trigonometry and series to solve problems.
- H6 uses the derivative to determine the features of the graph of a function.
- H7 uses the features of a graph to deduce information about the derivative.
- H8 uses techniques of integration to calculate areas and volumes.
- H9 communicates using mathematical language, notation, diagrams and graphs.



GIRRAWEE HIGH SCHOOL

HALF YEARLY EXAMINATION

2014

MATHEMATICS

*Time allowed - Two hours
(Plus 5 minutes' reading time)*

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Standard integrals are on laminated sheet provided.
- Board-approved calculators may be used.
- Answers to multiple choice questions should be written on your working paper.
- Each of questions 11-16 is to be returned on a *separate* piece of paper clearly marked Question 11 , Question 12 , etc. Each piece of paper must show your name.
- You may ask for extra pieces of paper if you need them.

PART A: Questions 1 – 10 (Multiple Choice) Write the letter corresponding to the correct answer on your working pages.

Question 1

$$\sum_{n=1}^{15} (5n - 3) =$$

- (A) 72 (B) 77 (C) 483 (D) 555

Question 2 The limiting sum of a series with common ratio $\frac{4}{5}$ is 15. The THIRD term of the series is

- (A) 3 (B) $\frac{12}{5}$ (C) $\frac{48}{25}$ (D) $\frac{192}{125}$

Question 3 The probability of selecting a red King from a standard deck of cards (4 of each kind, 13 of each suit, 52 in total) is

- (A) $\frac{1}{52}$ (B) $\frac{1}{26}$ (C) $\frac{1}{13}$ (D) $\frac{1}{4}$

Question 4 (2,7) is the midpoint of (–1,5) and

- (A) $(\frac{1}{2}, 6)$ (B) (5,9) (C) (–4,3) (D) $(\frac{3}{2}, 3)$

Question 5 The derivative of $y = \ln(x^2 + 1)$ is

- (A) $\frac{1}{x^2+1}$ (B) $\frac{x}{x^2+1}$ (C) $\frac{2x}{x^2+1}$ (D) $\frac{x^2}{x^2+1}$

Question 6 The derivative of $y = e^{-x^2}$ is

- (A) e^{-x^2} (B) e^{-2x} (C) $-x^2 e^{-x^2}$ (D) $-2x e^{-x^2}$

Question 7 $\int \sqrt{5x - 1} \cdot dx =$

- (A) $\frac{1}{5} (5x - 1) \sqrt{5x - 1}$ (B) $\frac{15}{2} (5x - 1) \sqrt{5x - 1}$ (C) $\frac{2}{15} \sqrt{5x - 1}$ (D) $\frac{2}{15} (5x - 1) \sqrt{5x - 1}$

Question 8 The quadratic equation $-2x^2 + 4x = 11$ has

- (A) No real roots (B) Two equal roots (C) Rational roots (D) Irrational roots

Question 9 A parabola has focus (0,0) and directrix $y = 4$. Its equation is

- (A) $x^2 = 8(2 - y)$ (B) $y^2 = 8(x - 2)$ (C) $x^2 = 8(y - 2)$ (D) $y^2 = 8(2 + x)$

Question 10 $y = \ln(3x - 1)$ has a vertical asymptote at

(A) $x = 0$

(B) $x = 1$

(C) $x = \frac{1}{3}$

(D) $x = 3$

PART B: Show all necessary working on your answer pages.

Question 11 (15 Marks)

Marks

(a) Find $\frac{dy}{dx}$ if

(i) $y = x^2 e^{-x}$

2

(ii) $y = \frac{\ln x}{x^2 + 1}$

2

(iii) $y = (\ln x)^3$

3

(iv) $y = \sqrt[3]{e^{2x}}$

2

(b) Find $\int_0^1 e^{3x} - 2x \cdot dx$

3

(c) Differentiate $y = e^{x^3}$. Hence find $\int x^2 e^{x^3} \cdot dx$

3

Question 12 (15 Marks)

(a) From a standard deck of cards (4 of each kind, 13 of each suit, 52 cards altogether) three cards are drawn *without replacement* to see how many Kings are drawn.

(i) Draw a probability tree of this experiment.

3

(ii) Find the probability of drawing two Kings.

2

(iii) Find the probability of drawing *at least* two Kings.

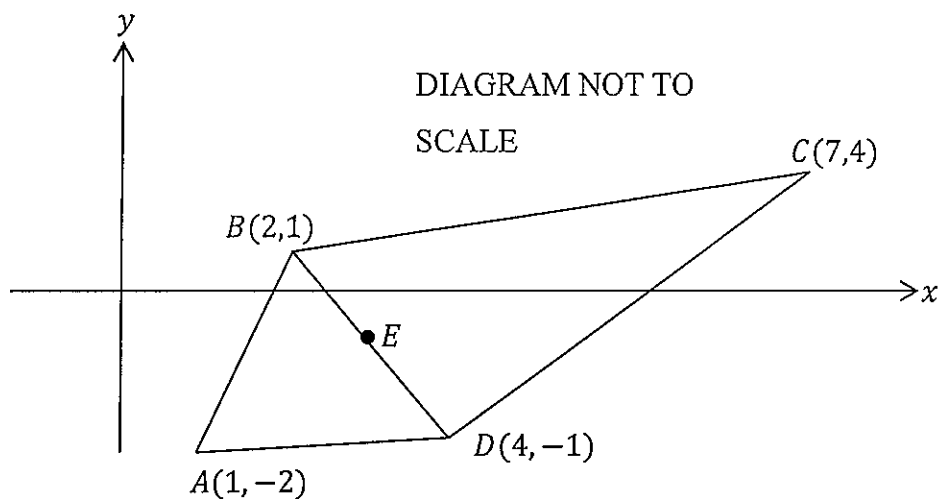
2

Question 12 continues on the next page

Question 12 (continued)**Marks**

(b) $A(1, -2)$, $B(2, 1)$, $C(7, 4)$ and $D(4, -1)$ are points on the number plane.

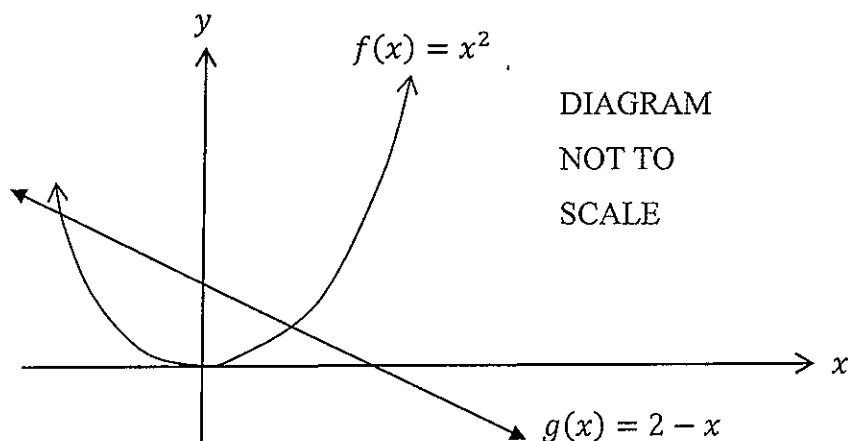
E is the midpoint of BD . (See diagram.)



- | | |
|--|---|
| (i) Find the equation of the line AC . | 2 |
| (ii) Show that $AC \perp BD$. | 2 |
| (iii) Find the coordinates of E and show that it lies on the line AC . | 3 |
| (iv) Hence or otherwise show that $ABCD$ is a kite. | 1 |

Question 13 (18 marks)

- (a) Find the locus of the set of points that are equally distant from the point $(1, -2)$ and the line $y = 4$. 3
- (b) On the graph below are graphs (not to scale) of the functions $f(x) = x^2$ and $g(x) = 2 - x$.



- | | |
|---|---|
| (i) Find the x coordinates of the points of intersection of the two curves. | 2 |
| (ii) Find the area enclosed by the two curves. | 3 |

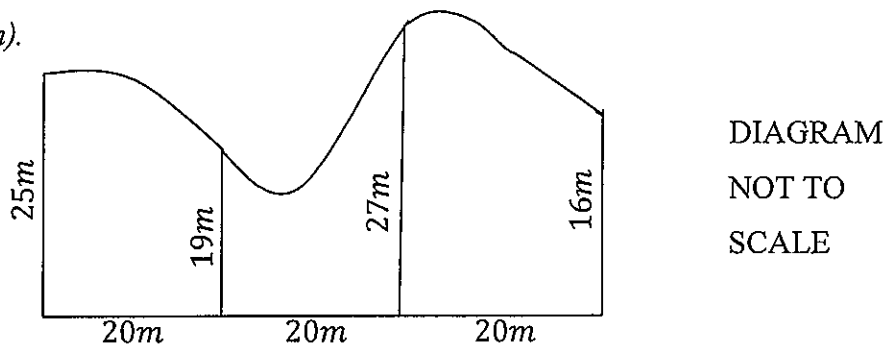
Question 13 continues on the next page

Question 13 (continued)	Marks
(c) Find the equation of the tangent to $y = 3e^{2x}$ at the point where $x = 1$.	3
(d) Stephanie is training for a swimming race. She swims 2km in the first week and each week she swims 300m more than the previous week.	
(i) How far does she swim in the tenth week?	2
(ii) How many weeks is it before she swims 10km in a week?	2
(iii) How many weeks will it be before she has accumulated at least 50km of training?	3

Question 14 (15 Marks)

(a) A surveyor wishes to find the approximate area of this beachside property.

(See diagram).



Find the approximate area using three application of the trapezoidal rule.	3
(b) (i) Use two application of Simpson's Rule (four sub intervals) to find the area enclosed by the curve $y = 2x^2 + x - 1$, the x axis and the lines $x = 1$ and $x = 3$.	3
(ii) The area worked out using Simpson's Rule is exactly the same as the area which would be found using integration. Why is this the case?	1
(c) Find the volume of the solid of revolution formed by rotating the curve $y = \ln(x)$ about the y axis between $y = 1$ and $y = 3$.	4
(d) If $y = 5e^{3x} - 2e^{-x}$, show that $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 0$.	4

Question 15 (15 Marks)

(a) (i) If $y = xe^x$ find $\frac{dy}{dx}$.	2
(ii) Hence find $\int xe^x \cdot dx$	2

Question 15 continues on the next page

Question 15 (continued)

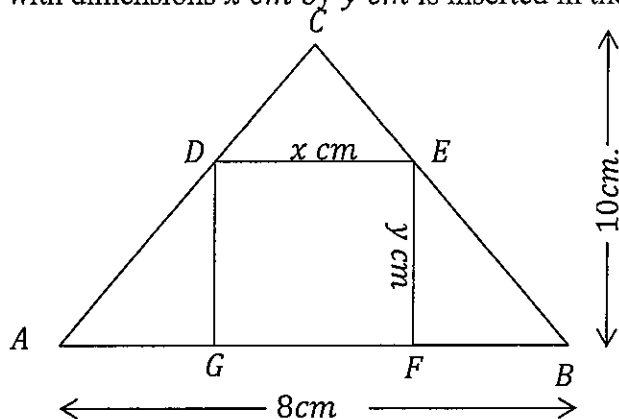
Marks

(b) For the curve $y = e^{-x^2}$

- | | |
|---|---|
| (i) Explain why $y = e^{-x^2}$ has no x intercepts. | 1 |
| (ii) Find the y intercept of $y = e^{-x^2}$ | 1 |
| (iii) Find any stationary points for $y = e^{-x^2}$ and determine their nature. | 4 |
| (iv) Find any points of inflexion for $y = e^{-x^2}$. | 3 |
| (v) Sketch the graph of $y = e^{-x^2}$ showing all relevant features. | 2 |

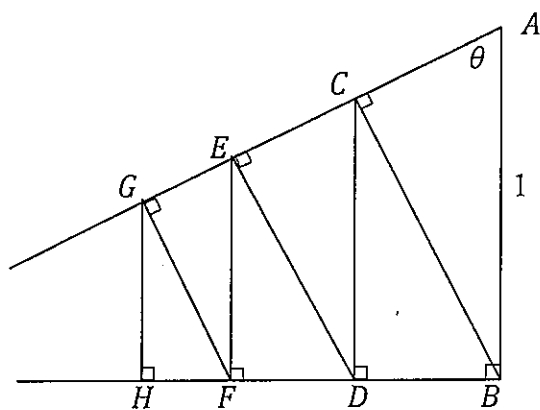
Question 16 (12 marks)

(a) A scalene triangle has a breadth of 8cm and a height of 10cm . A rectangle with dimensions $x\text{ cm}$ by $y\text{ cm}$ is inserted in the triangle as shown.



- | | |
|--|---|
| (i) Show that the area of the triangle ABC in terms of x and y is given by $\text{Area} = \frac{1}{2}x(10 - y) + \frac{1}{2}y(8 - x) + xy$ | 2 |
| (ii) Show that $y = \frac{1}{4}(40 - 5x)$ | 2 |
| (iii) Find the maximum area of the rectangle. | 4 |

(b) On the diagram below, $AB = 1$ and $\angle CAB = \theta$ as shown.



- | | |
|---|---|
| (i) Show that $CB = \sin\theta$ and $CD = \sin^2\theta$ | 2 |
| (ii) Show that the total distance $AB + BC + CD + DE + EF + FG + GH + HI + \dots = \sec\theta(\sec\theta + \tan\theta)$ | 2 |

HERE ENDETH THE EXAMINATION!

(1) D (2) C (3) B (4) B (5) C (6) D (7) D (8) A (9) A
(10) C

Q. (11) (a) (i) $y = x^{2-x}$

$$\begin{aligned}\frac{dy}{dx} &= u'v + v'u \\ &= 2xe^{-x} - e^{-x} \times x^2 \\ &= 2xe^{-x} - x^2e^{-x} \\ \text{or } &= \underline{x e^{-x} (2-x)}\end{aligned}$$

(ii) $y = \frac{\ln x}{x^2+1}$

$$\frac{dy}{dx} = \frac{u'v - v'u}{v^2}$$

$$= \frac{\frac{1}{x}(x^2+1) - 2x \ln x}{(x^2+1)^2}$$

$$= \frac{x^2+1 - 2x^2 \ln x}{x(x^2+1)^2}$$

(iii) $y = [\ln x]^3$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$= 3[\ln x]^2 \times \frac{1}{x}$$

$$= \frac{3[\ln x]^2}{x}$$

Q. (11)(a)(iv) $y = \sqrt[3]{e^{2x}}$

$$\therefore y = e^{\frac{2}{3}x}$$

$$\frac{dy}{dx} = \frac{2}{3} e^{\frac{2}{3}x}$$

(b) $\int_0^1 e^{3x} - 2x \cdot dx$

$$= \left[\frac{1}{3} e^{3x} - x^2 \right]_0^1$$

$$= \left(\frac{1}{3} \times e^{3 \times 1} - 1^2 \right) - \left(\frac{1}{3} \times e^{3 \times 0} - 0^2 \right)$$

$$= \frac{e^3}{3} - 1 - \frac{1}{3}$$

$$= \frac{e^3 - 4}{3}$$

$$\approx 5.36 \text{ (2DP)}$$

(c) $y = e^{x^3}$

By the chain rule

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$= e^x \times 3x^2$$

$$= 3x^2 e^x$$

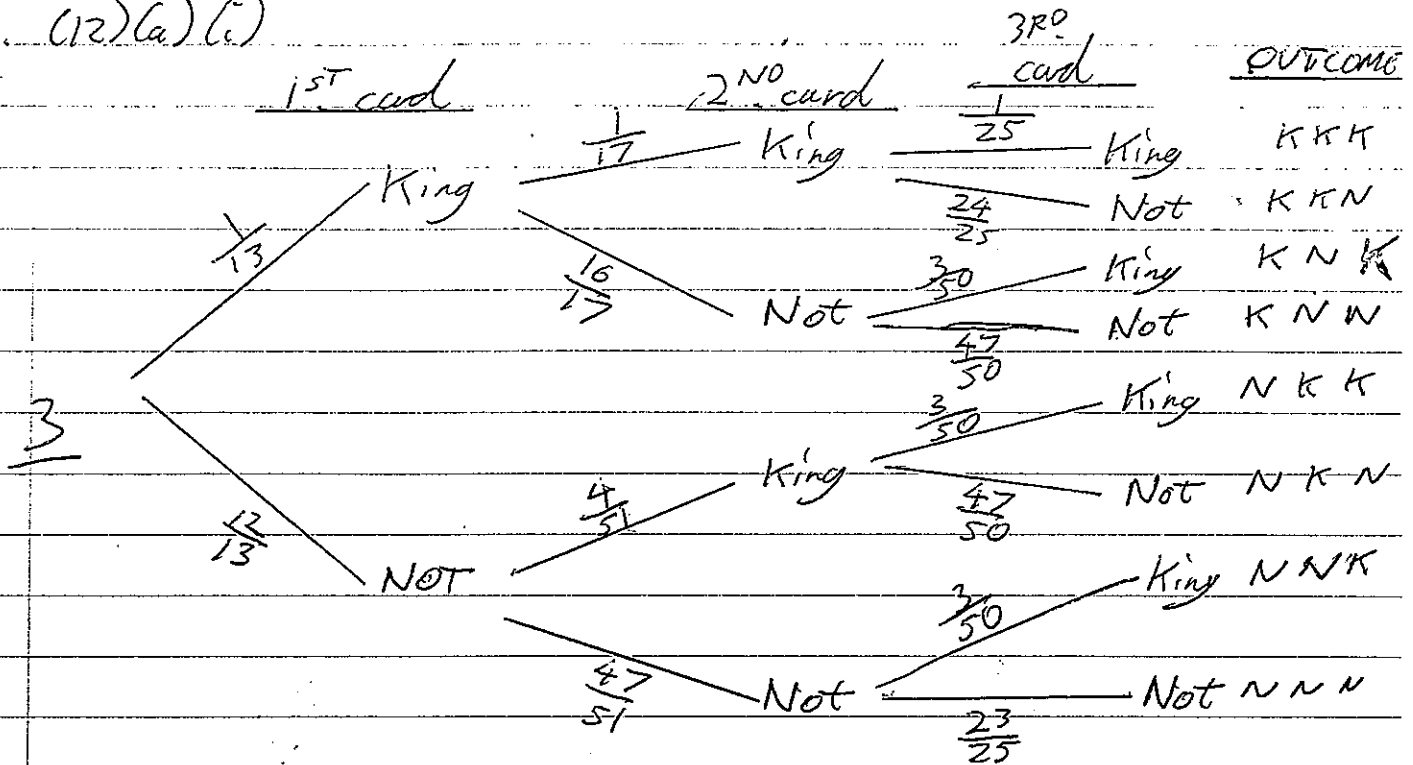
Hence $\int x^2 e^{x^3} \cdot dx$

$$= \frac{1}{3} \int 3x^2 e^{x^3} \cdot dx$$

$$= \frac{1}{3} e^{x^3} + C$$

p. 3

Q. (12)(a)(i)



(ii) $P(2 \text{ Kings})$

$$= \frac{1}{13} \times \frac{1}{17} \times \frac{24}{25} + \frac{1}{13} \times \frac{16}{17} \times \frac{3}{50} + \frac{12}{13} \times \frac{4}{51} \times \frac{3}{50}$$

$$+ \frac{12}{13} \times \frac{47}{51} \times \frac{3}{50}$$

$$= \frac{72}{5525}$$

(iii) $P(3 \text{ kings}) = \frac{1}{13} \times \frac{1}{17} \times \frac{1}{25}$

$$= \frac{1}{5525}$$

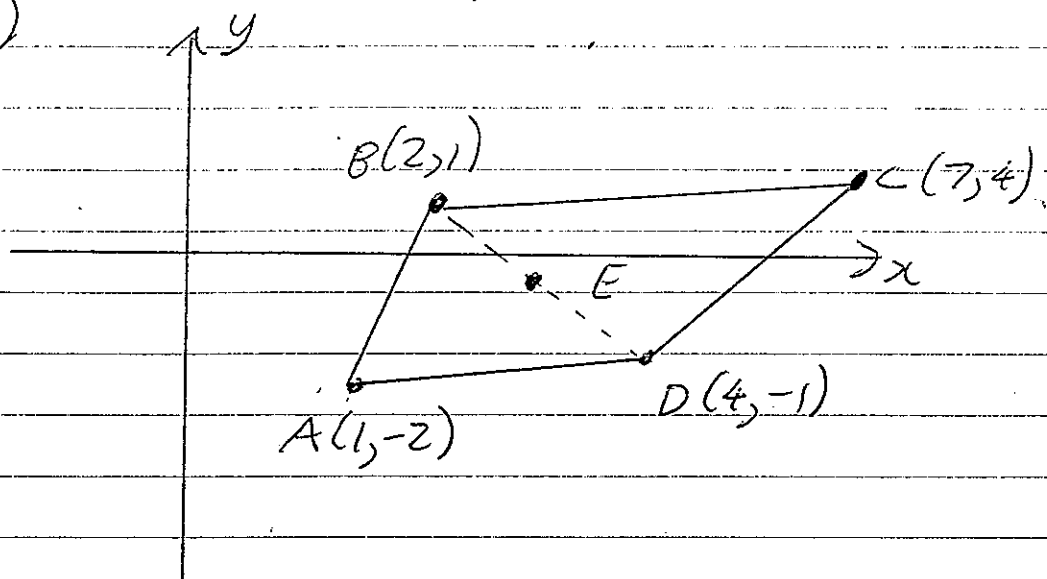
Total Probability of at least 2 Kings

$$= \text{Pr}(2 \text{ Kings}) + \text{Pr}(3 \text{ Kings})$$

$$= \frac{72}{5525} + \frac{1}{5525}$$

$$= \frac{73}{5525}$$

Q. (12)(b)



(ii) Equation of line AC:

$$\begin{aligned}
 m_{AC} &= \frac{y_2 - y_1}{x_2 - x_1} \\
 &= \frac{4 - (-2)}{7 - 1} \\
 &= 1.
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Equation of AC is } y - y_1 &= m(x - x_1) \quad \underline{2} \\
 y + 2 &= 1(x - 1) \\
 y + 2 &= x - 1 \\
 \underline{y} &= \underline{x - 3.}
 \end{aligned}$$

$$\begin{aligned}
 (\text{iii}) m_{BD} &= \frac{y_2 - y_1}{x_2 - x_1} \\
 &= \frac{-1 - 1}{4 - 2} \\
 &= -1.
 \end{aligned}$$

$$\begin{aligned}
 m_{AC} \times m_{BD} &= 1 \times -1 \\
 &= -1.
 \end{aligned}$$

$$\therefore \underline{AC \perp BD.}$$

PTO $\rightarrow$

Q. (12)(b) [continued]:

(iii) E is midpoint of BD

$$= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$= \left(\frac{2+4}{2}, \frac{1-1}{2} \right)$$

$$E = (3, 0).$$

Show E lies on AC:

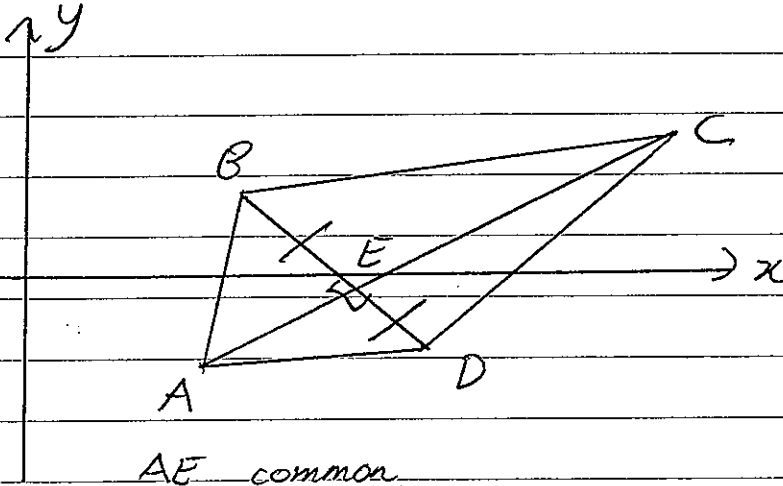
$$y = x - 3$$

$$0 = 3 - 3$$

True $\Rightarrow$ E lies on AC.

(iv) As diagonal AC bisects diagonal BD at right angles, ABCD is a kite. 1 Sufficient.

For more detail:



AE common.

$$\triangle ABE \cong \triangle ADE \text{ [SAS]}.$$

$$\therefore AB = AD \text{ [matching sides, } \triangle ABE \cong \triangle ADE].$$

$$\angle BAE = \angle DAE \text{ [" " " "]}.$$

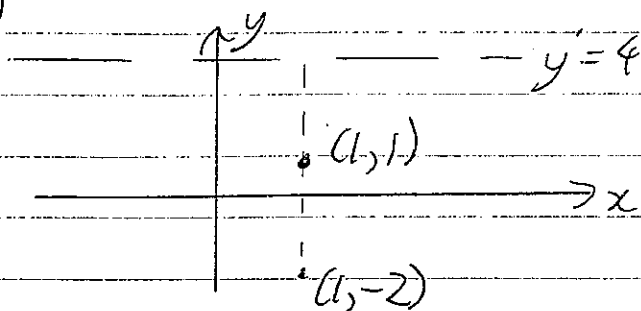
$$\therefore \triangle ABC \cong \triangle ADC \text{ [SAS]}.$$

$$BC = DC \text{ [matching sides, } \triangle ABC \cong \triangle ADC].$$

$\therefore$ ABCD is a kite.

p.6

Q. 13)(a)

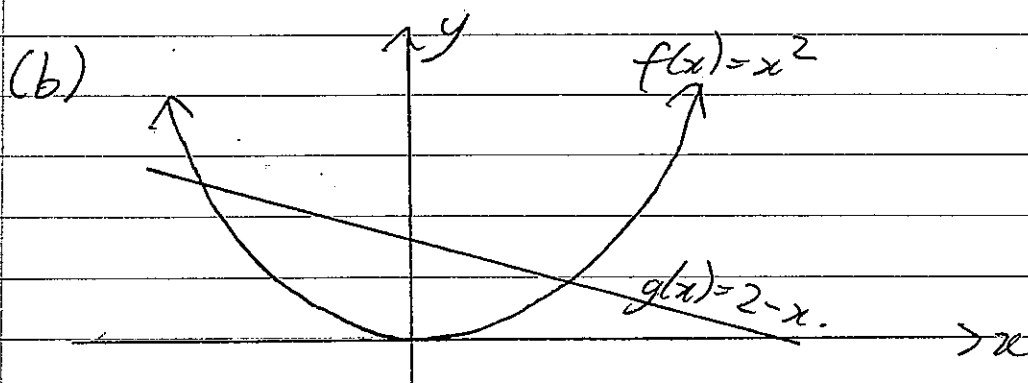


Note: Equidistant from POINT [S] & LINE [D]

 $\therefore$ Parabola
 $S = (1, -2)$ directrix $= y = 4$. Vertex $= (1, 1)$
 $a = -3$.

$$\therefore \text{By } (x-p)^2 = 4a(y-q)$$

$$(x-1)^2 = -12(y-1) \quad | \quad \underline{3}$$



$$(i) \quad x^2 = 2 - x \quad |$$

$$x^2 + x - 2 = 0$$

$$(x+2)(x-1) = 0$$

$$x = -2 \text{ or } x = 1 \quad | \quad \underline{2}$$

$$(ii) \text{ Area} = \int_{-2}^1 (2-x) - x^2 \, dx$$

$$= \left[2x - \frac{1}{2}x^2 - \frac{1}{3}x^3 \right]_{-2}^1 \quad |$$

$$= \left(2 \times 1 - \frac{1}{2} \times 1^2 - \frac{1}{3} \times 1^3 \right) - \left(2 \times (-2) - \frac{1}{2} \times (-2)^2 - \frac{1}{3} \times (-2)^3 \right) \quad | \quad \underline{3}$$

$$= \frac{7}{6} - \left(-\frac{10}{3} \right) = \frac{9}{2} \text{ square units} \quad |$$

p. 7

Q. (13)(c) $y = 3e^{2x}$

Where $x = 1$, $y = 3e^{2 \times 1}$
 $= 3e^2$

m of tangent: $\frac{dy}{dx} = 6e^{2x}$

At $x = 1$, $\frac{dy}{dx} = 6e^{2 \times 1}$
 $= 6e^2$

$y - y_1 = m(x - x_1)$

$y - 3e^2 = 6e^2(x - 1)$

$y - 3e^2 = 6e^2x - 6e^2$

$y = 6e^2x - 3e^2$

OR $6e^2x - y - 3e^2 = 0$

(d)(i) Stephanie is swimming

$2000 + 2300 + 2600 + \dots$

Arithmetic progression: $a = 2000$ $d = 300$

By $T_n = a + (n-1)d$

10th week $= 2000 + 9 \times 300$

$= 4700m$

(ii) By $T_n = a + (n-1)d$

$10000 = 2000 + (n-1) \times 300$

$8000 = 300n - 300$

$8300 = 300n$

$27\frac{2}{3} = n$

It will be 28 weeks before she breaks 10km in a week.

(iii) By $S_n = \frac{n}{2} [2a + (n-1)d]$

$50000 = \frac{n}{2} [2 \times 2000 + (n-1) \times 300]$

$50000 = 2000n + 150n^2 - 150n$

PTO $\Rightarrow$

p. 8

Q. (13)(d)(iii) [continued].

$$150n^2 + 1850n - 50\,000 = 0$$

$$3n^2 + 37n - 1000 = 0$$

$$\text{By } n = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$n = \frac{-37 \pm \sqrt{37^2 - 4 \times 3 \times -1000}}{2 \times 3}$$

$$n = \frac{-37 - \sqrt{13269}}{6} \text{ or } n = \frac{-37 + \sqrt{13269}}{6}$$

$$n = -6.76\ldots \text{ or } n = 13.07\ldots$$

$\rightarrow$ No negative weeks

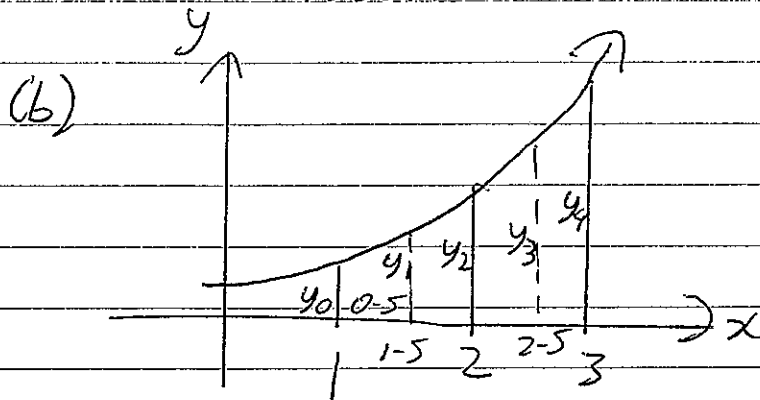
|

$\therefore$ It will be 14 weeks before Stephanie accumulates 50km of training.

Q. (14) (a) $A \doteq \frac{h}{2} (y_0 + 2(y_1 + y_2) + y_3)$ or appropriate table or other way of setting out trapezoidal rule.

$$= \frac{20}{2} (25 + 2 \times (19 + 27) + 16)$$

$$= 1330 \text{ m}^2$$



$$h = 0.5 \quad y_0 = 2 \times 1^2 + 1 - 1 = 2$$

$$y_1 = 2 \times 1.5^2 + 1.5 - 1 = 5$$

$$y_2 = 2 \times 2^2 + 2 - 1 = 9$$

$$y_3 = 2 \times 2.5^2 + 2.5 - 1 = 14$$

$$y_4 = 2 \times 3^2 + 3 - 1 = 20$$

$$A = \frac{h}{3} (y_0 + y_4 + 4(y_1 + y_3) + 2y_2)$$

$$= \frac{0.5}{3} [2 + 20 + 4(5 + 14) + 2 \times 9]$$

$$= \frac{58}{3} \text{ square units.}$$

(ii) Simpson's Rule is based on approximating the curve using a parabola. As $y = 2x^2 + x - 1$ is a parabola it will be completely accurate.

PTO $\rightarrow$

p. 10

Q. (14)(c) $y = \ln x$ about y axis
 $\therefore x = e^y$

$$V = \pi \int_{y=a}^{y=b} x^2 dy$$

$$= \pi \int_1^3 (e^y)^2 dy$$

$$= \pi \int_1^3 e^{2y} dy$$

$$= \pi \left[\frac{1}{2} e^{2y} \right]_1^3$$

$$= \frac{\pi}{2} [e^6 - e^2]$$

$$\approx \underline{622.1 \text{ cubic units.}}$$

(d) If $y = 5e^{3x} - 2e^{-x}$,

$$\frac{dy}{dx} = 15e^{3x} + 2e^{-x}$$

$$\frac{d^2y}{dx^2} = 45e^{3x} - 2e^{-x}$$

LHS:

$$\therefore \frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y$$

$$= 45e^{3x} - 2e^{-x} - 2(15e^{3x} + 2e^{-x}) - 3(5e^{3x} - 2e^{-x})$$

$$= 45e^{3x} - 2e^{-x} - 30e^{3x} - 4e^{-x} - 15e^{3x} + 6e^{-x}$$

$$= 0$$

$$= \text{RHS Q.E.D.}$$

p. 11

Q. (15)(a)(i) $y = xe^x$

$$\begin{aligned}\frac{dy}{dx} &= u'v + v'u \\ &= e^x + e^x \cdot x \\ &= e^x + xe^x\end{aligned}$$

1 2

(ii) $\therefore$ As, if $y = xe^x$, $\frac{dy}{dx} = e^x + xe^x$

$$\therefore \int e^x + xe^x \cdot dx = xe^x$$

$$\int e^x \cdot dx + \int xe^x \cdot dx = xe^x$$

2

$$\int xe^x \cdot dx = xe^x - \int e^x \cdot dx$$

$$\int xe^x \cdot dx = xe^x - e^x + C.$$

(b) (i) $y = e^{-x^2} \Rightarrow e$ to the power of ANY real no. > 0

$$\therefore y > 0 \text{ for all real } x.$$

(ii) y intercept: $x = 0$

$$y = e^{-0^2}$$

$$= 1.$$

$$y \text{ intercept} = (0, 1)$$

(iii) $y = e^{-x^2}$

$$\frac{dy}{dx} = -2xe^{-x^2}$$

$$\frac{dy}{dx} = 0 \Rightarrow -2xe^{-x^2} = 0$$

$$\text{As } e^{-x^2} \neq 0, -2x = 0$$

$$x = 0$$

Nature of turning point: $\frac{d^2y}{dx^2} = -2e^{-x^2} + -2x \cdot -2xe^{-x^2}$

$$= (4x^2 - 2)e^{-x^2}$$

$$\text{At } x = 0, \frac{d^2y}{dx^2} = (4 \times 0^2 - 2)e^{-0^2}$$

$$= -2.$$

$$\rightarrow \text{Local MAXIMUM at } (0, 1)$$

[actually GLOBAL]

4

Q. (15)(b)(iv) Points of inflexion:

$$\frac{d^2y}{dx^2} = 0 \text{ \& changes sign.}$$

$$(4x^2 - 2)e^{-x^2} = 0$$

$$4x^2 - 2 = 0 \text{ [as } e^{-x^2} \neq 0].$$

$$4x^2 = 2$$

$$x^2 = \frac{1}{2}$$

$$x = \pm \frac{1}{\sqrt{2}}$$

$$\begin{aligned} \text{At } x = \pm \frac{1}{\sqrt{2}}, y &= e^{-\left(\frac{1}{\sqrt{2}}\right)^2} \\ &= e^{-\frac{1}{2}} \\ &= \frac{1}{\sqrt{e}}. \end{aligned}$$

Possible points of inflexion at $\left(\pm \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{e}}\right)$ 3

Noting that $\frac{d^2y}{dx^2} < 0$ for $-\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}}$
[from testing turning point].

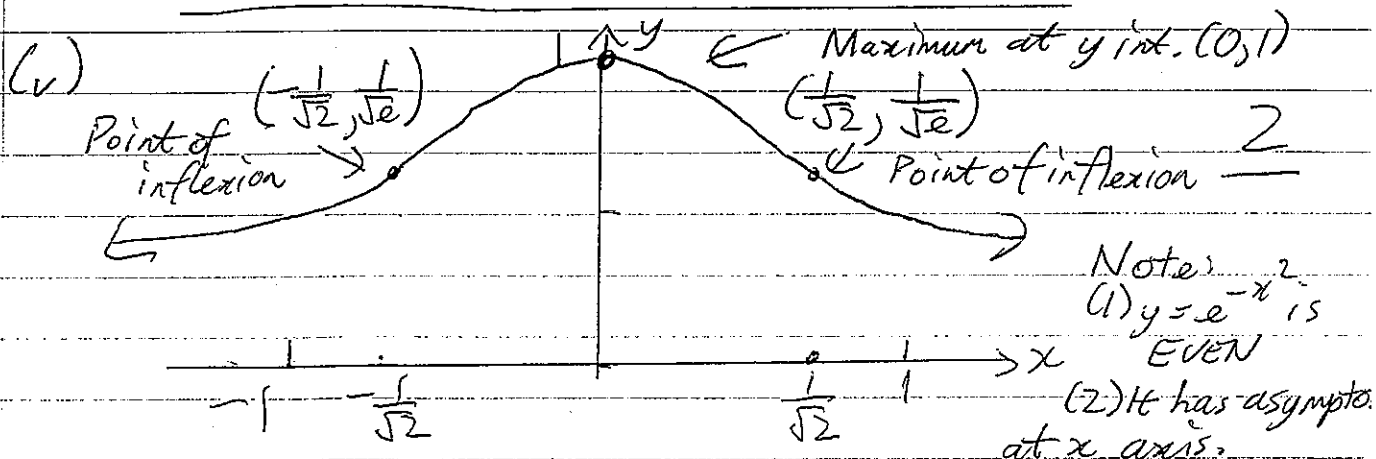
Testing left of $x = -\frac{1}{\sqrt{2}}$.

$$\begin{aligned} \text{At } x = -1, \frac{d^2y}{dx^2} &= (4x^2 - 2)e^{-x^2} \\ &= (4(-1)^2 - 2)e^{-1} \\ &= \frac{2}{e} > 0 \end{aligned}$$

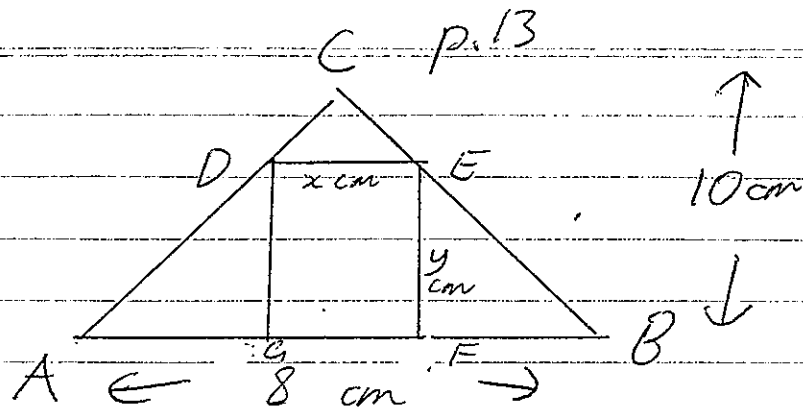
Testing right:

$$\begin{aligned} \text{At } x = 1, \frac{d^2y}{dx^2} &= (4x^2 - 2)e^{-x^2} \\ &= (4(1)^2 - 2)e^{-1} \\ &= \frac{2}{e} > 0. \end{aligned}$$

$\therefore$ Points of inflexion at $\left(\pm \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{e}}\right)$.



Q. (16)(a)



(i) Area top triangle $DEC = \frac{1}{2} Bh$

$$= \frac{1}{2} \times (10 - y) \quad |$$

Combined area $\triangle ADG$ and $\triangle EFB$

$$= \frac{1}{2} (8 - x) \times y \quad |$$

Area rectangle $DEFG = xy \quad |$

$\therefore$ Total area $\triangle ABC$

$$= \frac{1}{2} \times (10 - y) + \frac{1}{2} y (8 - x) + xy \quad |$$

(ii) Using Area $\triangle ABC = \frac{1}{2} Bh$

$$= \frac{1}{2} \times 8 \times 10$$

$$= 40 \text{ cm}^2$$

$$\frac{1}{2} \times (10 - y) + \frac{1}{2} y (8 - x) + xy = 40 \quad [\text{from (i)}] \quad |$$

$$5x - \frac{1}{2} xy + 4y - \frac{1}{2} xy + xy = 40$$

$$5x + 4y = 40 \quad |$$

$$4y = 40 - 5x \quad |$$

$$y = \frac{1}{4} (40 - 5x) \quad |$$

(iii) Maximum rectangle area

$$\text{Area} = xy$$

$$A = \frac{1}{4} x (40 - 5x) \quad |$$

$$A = 10x - \frac{5}{4} x^2$$

PTO $\rightarrow$

Q. (16)(a)(iii) [continued]:

Finding maximum area.

$$\frac{dA}{dx} = 10 - \frac{5}{2}x.$$

Finding where $\frac{dA}{dx} = 0$

$$10 - \frac{5}{2}x = 0$$

$$x = 4.$$

Testing to see if maximum: $\frac{d^2A}{dx^2} = -\frac{5}{2}$ Maximum area when $x = 4$.

$$\text{Note: Could do } A = 10x - \frac{5}{4}x^2$$

$$= 10 \times 4 - \frac{5}{4} \times 4^2$$

$$= \underline{20 \text{ cm}^2}$$

$$\begin{aligned} \textcircled{\text{OR}} \quad y &= \frac{1}{4}(40 - 5x) \\ &= \frac{1}{4}(40 - 5 \times 4) \\ &= 5 \text{ cm.} \end{aligned}$$

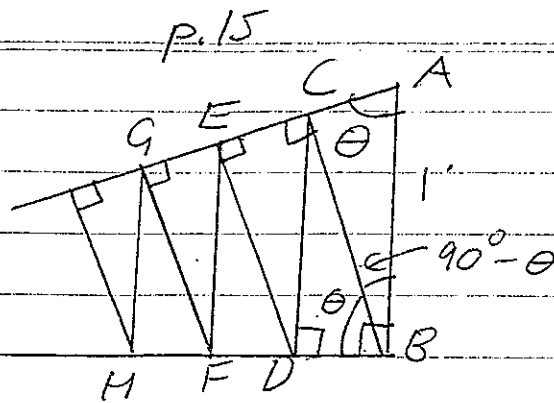
or max here.

4

$$\begin{aligned} \text{Area} &= xy \\ &= 4 \times 5 \\ &= \underline{20 \text{ cm}^2} \end{aligned}$$

Maximum area of rectangle = 20 cm^2

Q. (16)(b)



(i) Using $\triangle ABC$, $\sin \theta = \frac{CB}{AB}$

$= \frac{CB}{1}$

$\therefore \sin \theta = CB. \quad (1)$

$\angle CBA = 90^\circ - \theta$ [L sum $\triangle ABC$].

$\therefore \angle CBD = \theta$ [adjacent $\angle$'s].

Using $\triangle CBD$

$\sin \theta = \frac{CD}{CB}$

$\sin \theta = \frac{CD}{\sin \theta}$
x BS by $\sin \theta$

$\sin^2 \theta = CD.$

$\therefore CB = \sin \theta, CD = \sin^2 \theta.$

(ii) $AB + BC + CD + DE + \dots$

$= 1 + \sin \theta + \sin^2 \theta + \sin^3 \theta + \dots$

By the limiting sum of a GP, $a=1$ $r=\sin \theta$

$AB + BC + \dots = \frac{1}{1 - \sin \theta} \times \frac{1 + \sin \theta}{1 + \sin \theta}$

$= \frac{1 + \sin \theta}{1 - \sin^2 \theta}$

$= \frac{1 + \sin \theta}{\cos^2 \theta}$

$= \frac{1}{\cos^2 \theta} + \frac{\sin \theta}{\cos^2 \theta}$

$= \frac{1}{\cos^2 \theta} + \frac{1}{\cos \theta} \times \frac{\sin \theta}{\cos \theta}$

$= \sec^2 \theta + \sec \theta \tan \theta$

$= \sec \theta (\sec \theta + \tan \theta)$

$= \text{RHS QED.}$

END OF SOLUTIONS!